

(a) Alert at period 13

“The supply manager reports that routine supplier turnaround on the Rotterdam lane has become noticeably longer, rather than a particular consignment being temporarily held.”

SHOCKSPEC Gemini 3.8 Flash

shock_family	lead_time_shift
target_stream	arrival
direction	arrival_delayed
magnitude_bin	medium
onset_window	[0, 2]
persistence	persistent

the model never writes an order or a threshold

the hidden truth
lead time +1 period, from period 14

(b) Registered gate (arm 10)

waits for the threshold, then acts all at once on the stated size

the stated hypothesis only

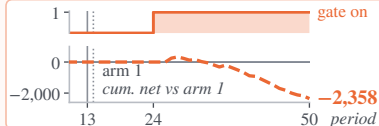
■ lead-time shift

magnitude medium

all at once
 $E_{j,t}$ after τ_j
switch at $E_{j,t} \geq 1/\alpha_j$

stated size, compiled

+2 periods ✗
true shift: +1 period
order from the compiled configuration



hypotheses

commitment

size and order

the example episode

alert onset

(c) Certify-then-hedge

prices each hypothesis by its evidence and hedges by that price; with nothing live, it orders as arm 1

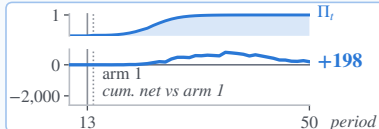
fixed $\mathcal{H}$ prior w_{jk}

■ demand up
■ demand down
■ lead-time shift
■ upward pulse
■ $\lambda/|\mathcal{H}|$ each
■ $+(1-\lambda)$ if named
 $\lambda = 0.25$: the answer only reweights $\mathcal{H}$ and dates $\hat{k}_j$

graded
 $E_{jk,t}$ for every k
 $\pi_{jk,t} \propto a_j w_{jk} E_{jk,t-1}$

sized from the data

+1 period ✓
true shift: +1 period
order: $p/(p+h)$ quantile of the mixture



■ $\mathbb{E}_0[\Pi_\sigma] \leq A$
at every stopping time σ

when nothing has changed,
whatever the model says

if the null and arrival law are known;
as run, up to 2.7A (Limitations)